

MALMSTROM AFB, MT, USA

WMO: 727755

Lat: 47.517N

Lon: 111.183W

Elev: 3472

StdP: 12.94

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24112

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-14.2	-8.8	-19.9	2.1	-12.9	-14.3	2.9	-8.7	34.0	44.0	31.2	43.0	5.9	240	0.665

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.0	94.2	62.4	90.6	61.1	87.2	60.0	65.5	86.4	63.4	83.9	61.7	82.2	10.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.5	83.2	71.7	56.0	75.9	69.4	54.0	70.5	68.1	32.1	86.1	30.4	84.2	29.1	81.8	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.8	27.0	24.4	DB	-23.7	99.3	7.1	3.8	-28.8	102.1	-33.0	104.3	-36.9	106.5	-42.1	109.2	(4)
(5)			WB	-23.7	68.7	6.8	3.6	-28.6	71.3	-32.6	73.4	-36.4	75.4	-41.4	78.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.6	27.1	26.0	36.4	44.5	53.3	62.1	70.8	68.9	59.8	46.6	35.5	27.3
	DBStd	19.52	16.35	16.97	13.46	9.11	8.04	6.88	6.50	6.81	8.81	10.14	14.17	15.37
	HDD50	3452	713	675	438	209	56	3	0	0	20	178	453	707
	HDD65	7162	1173	1092	888	616	370	134	21	41	199	571	887	1170
	CDD50	2226	4	4	16	43	158	366	644	585	315	73	16	2
	CDD65	460	0	0	0	1	8	46	199	160	45	1	0	0
	CDH74	6792	0	0	2	34	184	727	2799	2240	762	44	0	0
	CDH80	2956	0	0	0	2	40	237	1348	1030	292	7	0	0
	WSAvg	10.6	13.0	11.4	11.6	10.6	9.3	9.2	8.3	8.2	8.9	11.0	12.9	12.7
	PrecAvg	13.0	0.4	0.4	0.6	1.3	2.2	2.8	1.3	1.3	1.1	0.8	0.5	0.4
(15) Precipitation	PrecMax	21.2	1.1	1.2	1.4	2.4	4.3	5.3	5.8	2.6	2.0	2.2	1.2	1.2
	PrecMin	8.7	0.0	0.0	0.0	0.3	0.3	1.1	0.1	0.2	0.1	0.1	0.0	0.0
	PrecStd	2.9	0.2	0.3	0.4	0.6	0.9	1.3	1.3	0.7	0.5	0.5	0.3	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.1	62.8	70.3	78.9	85.1	92.2	100.5	97.7	93.1	79.5	69.1	59.5
		MCWB	44.3	45.5	49.3	54.0	59.1	63.3	69.0	63.1	60.7	55.8	50.0	43.3
	2%	DB	55.0	56.8	64.0	71.9	79.1	86.5	95.4	93.4	87.7	72.9	61.7	54.0
		MCWB	41.5	42.2	45.9	51.7	57.1	61.9	63.5	61.4	58.3	52.5	46.1	41.3
	5%	DB	50.8	52.2	59.6	66.0	74.5	81.8	91.4	90.0	83.0	66.9	57.6	49.8
		MCWB	39.0	40.1	43.7	48.2	55.0	59.7	62.0	60.1	57.1	50.1	44.1	38.7
	10%	DB	47.2	47.8	55.1	60.8	69.9	77.8	87.8	86.1	77.7	62.2	54.0	45.9
		MCWB	37.2	38.0	41.6	45.5	52.8	58.1	60.8	59.3	55.3	47.9	42.2	36.5
	0.4%	WB	46.1	46.9	50.2	55.6	61.3	67.5	71.3	65.5	63.1	57.0	51.2	45.3
		MCDB	58.6	60.0	67.8	73.3	83.3	83.6	97.1	90.6	84.9	75.8	66.5	58.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.9	43.3	47.2	52.0	58.3	64.1	66.6	63.2	60.4	53.8	47.6	41.9
		MCDB	53.8	55.1	62.3	70.1	75.4	81.3	89.0	86.9	82.5	69.7	60.3	52.4
	5%	WB	39.7	40.7	44.6	49.2	56.2	61.8	64.3	61.7	58.4	51.1	44.9	39.3
		MCDB	49.8	51.5	58.2	64.4	71.5	77.9	85.0	84.7	79.4	64.8	56.6	48.6
	10%	WB	37.5	38.0	42.0	46.3	54.1	59.6	62.3	60.3	56.3	48.5	42.5	36.8
		MCDB	46.2	47.6	54.1	59.4	67.7	74.7	82.7	81.9	75.8	60.9	53.1	45.3
(35) Mean Daily Temperature Range	5% DB	MDBR	19.1	19.8	21.7	22.0	22.6	24.5	29.0	28.6	25.8	21.4	19.1	18.1
		MCDBR	21.5	24.3	27.3	31.5	30.9	31.5	34.2	35.0	35.0	29.1	23.4	21.0
		MCWBR	13.0	14.3	14.7	15.6	13.9	13.0	11.9	12.6	14.4	13.9	13.0	12.9
	5% WB	MCDBR	20.4	23.5	26.2	30.1	28.1	28.9	31.3	31.4	32.0	26.9	22.8	20.1
		MCWBR	13.0	14.6	14.6	15.3	13.4	13.5	12.4	11.9	13.6	13.6	13.5	12.7
(40) Clear-Sky Solar Irradiance	taub		0.251	0.259	0.275	0.321	0.331	0.332	0.335	0.355	0.316	0.286	0.258	0.245
	taud		2.505	2.516	2.537	2.404	2.424	2.464	2.470	2.374	2.459	2.545	2.533	2.494
	Ebn at Noon		274	295	305	296	296	295	292	281	285	280	269	262
	Edh at Noon		19	23	27	34	35	34	33	35	29	23	18	17
(44) All-Sky Solar Radiation	RadAvg		460	771	1150	1535	1816	2094	2289	1890	1361	862	511	366
	RadStd		29	53	91	87	162	140	116	105	103	75	42	26

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page